

What is Spring Boot Actuator?

Spring Boot Actuator provides **production-ready features** to **monitor and manage** Spring Boot applications using **HTTP endpoints or JMX**.

Purpose of Actuator

- Monitor application **health**
- Track **performance metrics**
- Inspect **configuration & beans**
- Manage **logging**
- Support **production debugging**

Core Endpoints (Most Used)

Endpoint	URL
All endpoints list	/actuator
Health	/actuator/health
Health details	/actuator/health/db
Info	/actuator/info
Metrics	/actuator/metrics
Environment	/actuator/env
Beans	/actuator/beans
Config properties	/actuator/configprops
Conditions	/actuator/conditions
Loggers	/actuator/loggers

Mappings	/actuator/mappings
Scheduled tasks	/actuator/scheduledtasks
Caches	/actuator/caches

Metrics & Monitoring

Endpoint	URL
All metrics	/actuator/metrics
JVM memory	/actuator/metrics/jvm.memory.used
JVM threads	/actuator/metrics/jvm.threads.live
HTTP requests	/actuator/metrics/http.server.requests
Process uptime	/actuator/metrics/process.uptime
System CPU	/actuator/metrics/system.cpu.usage

CORE Actuator Endpoints – Purpose

These endpoints help you **understand application configuration, status, and behavior**.

Endpoint	Purpose	When to Use
/health	Shows application health (UP/DOWN)	Monitoring, load balancers
/info	App info (name, version, build)	App identification
/env	Environment variables & properties	Debug config issues
/beans	All Spring beans loaded	Debug dependency injection
/configprops	Configuration properties	Verify property binding
/conditions	Auto-configuration report	Debug startup issues
/mappings	All request mappings	Debug API routing
/loggers	View/change log levels	Runtime logging control
/caches	Cache info	Cache debugging

<code>/scheduledtasks</code>	Scheduled jobs info	Verify cron jobs
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CORE = How the app is built & configured

METRICS Actuator Endpoints – Purpose

Metrics endpoints provide **performance, resource, and runtime statistics**.

Endpoint	Purpose	Example Use
<code>/metrics</code>	List all metrics	Explore available metrics
<code>/metrics/jvm.memory.used</code>	JVM memory usage	Memory leak detection
<code>/metrics/jvm.threads.live</code>	Live threads count	Thread issue analysis
<code>/metrics/http.server.requests</code>	HTTP request stats	API performance
<code>/metrics/process.cpu.usage</code>	CPU usage	Load monitoring
<code>/metrics/system.cpu.usage</code>	System CPU usage	Server health
<code>/metrics/process.uptime</code>	App uptime	Reliability tracking
<code>/metrics/hikaricp.connections.active</code>	DB connections	DB performance

METRICS = How the app is performing

If Some URLs Don't Work

That means:

- Endpoint not exposed
- Disabled by default
- Blocked by Spring Security